Exercise 3

A tiled room

At the end, `MovementLab` is a small room painted on a tilemap: a grass floor between two crate walls.

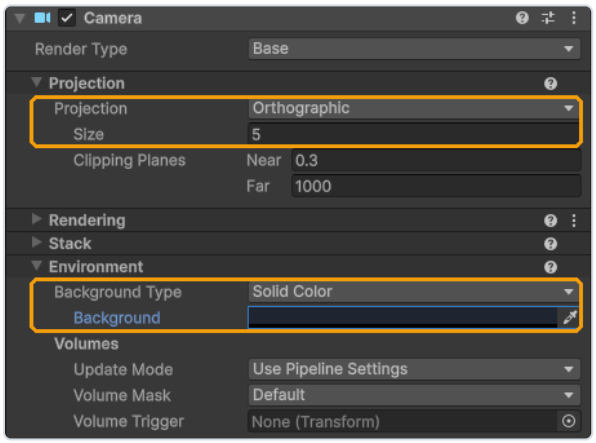
Before you start

- Exercise 2 done and pushed: the tiles, the character sheet and the sorting layers. In a new session, run `git pull` first

1 A scratch scene

`MovementLab` is a bare room for testing movement, away from the level you build in the exercise.

- **File** › **New Scene**, pick **Basic 2D (Built-in)**, **Create**. Not the pinned Lit 2D (URP): its Global Light 2D lights only the `Default` sorting layer, so the tiles would render black.
- `Ctrl` + `S`, and save it as `MovementLab` in `Assets/Scenes`.
- Select `Main Camera`: Projection **Orthographic** and Size `5` are already set, which shows 10 units of height.
- Under Environment, set Background Type to **Solid Color** (the template uses Skybox) and pick a dark Background.



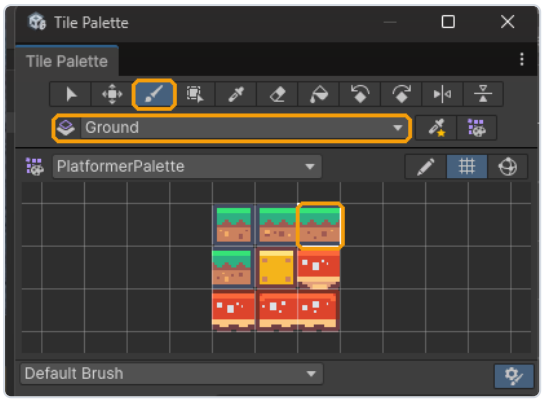
Main Camera: Orthographic, Size 5, a solid dark background.

2 A tilemap floor

A tilemap paints the level cell by cell from a palette; with PPU 18, each tile fills exactly one cell.

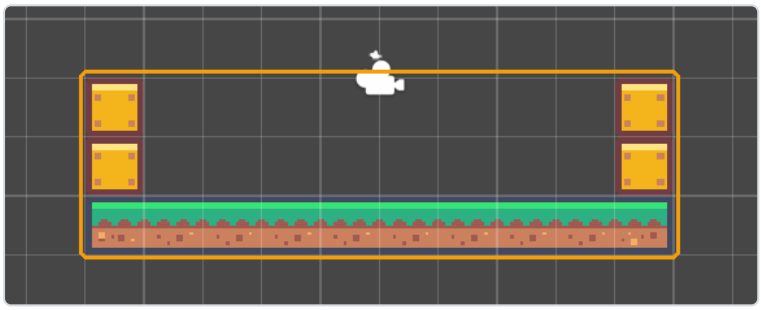
- **GameObject** › **2D Object** › **Tilemap** › **Rectangular**: a `Grid` with a `Tilemap` child.
- Rename the child `Ground` (`F2`) and set its Tilemap Renderer's Sorting Layer to `Level`.
- **Window** › **2D** › **Tile Palette**, then **Create New Tile Palette** from its dropdown: name it `PlatformerPalette`, **Create**, and save it in a new folder `Assets/Tiles`.
- In `Assets/Sprites/Tiles`, `Ctrl`-click `tile_0000` to `tile_0003`, `tile_0009` and `tile_0012` to `tile_0015`, drag them into the palette, and save the generated tiles in `Assets/Tiles`. Take them from `Tiles` itself: `Backgrounds` and `Characters` reuse the same file names but hold 24 px images, which cover 1.33 cells.

- Take the Brush (`B`), click the middle grass tile (`tile_0002`) and check that Active Tilemap reads `Ground`: paint always lands on the active tilemap.



Brush on, Active Tilemap on Ground, the grass tile framed; each tile is one cell.

- With Box Fill (`U`), drag a strip of 10 cells at y = -3; then brush the grass ends (`tile_0001`, `tile_0003`) onto its ends and a two-crate wall (`tile_0009`) on top of each end.



Ten cells at y = -3 and a wall at each end; the tile edges sit on the grid lines.

- `Ctrl` + `S`, then commit and push with the message `Add MovementLab with a tilemap floor`.